Proceedings of the 1998 IEEE
International Conference on Robotics & Automation
Leuven, Belgium • May 1998

A Hybrid Technique to Supply Indoor Service Robots

Chatchai WATTANASIN, Yasumichi AIYAMA, Daisuke KURABAYASHI,
Jun OTA and Tamio ARAI

Department of Precision Machinery Engineering
Graduate School of Engineering, The University of Tokyo
7-3-1 Hongo, Bunkyo-ku, Tokyo 113 Japan
chatchai@prince.pe.u-tokyo.ac.jp

Abstract

Service robots are extensively installed with end-effecters, which require high energy, to achieve a specific task. Robots require high-power energy sources that can meet several goals simultaneously. Given the technology of today a battery-driven robot cannot work for long on a single charge. To develop a new powerful battery technology may take many exhausted years. In this paper, we present a dual-energized scheme called HYbrid POwer Supply (HYPOS) that integrates the function of batteries and household electric system to supply a robot. This technique can be realized by the existing technology and is applicable to a variety of indoor applications. The paper includes the description of the prototype robot and the experiments that illustrate the effectiveness and practicalness of this proposed system.

1. Introduction

Robotic applications nowadays are being moved gradually to service tasks. Service robots perform a task that is not for productivity but act like "a substitute that would replace one like a servant" [1]. These robots are distinguished from industrial robots by many faculties, which permit effective and unobtrusive operation in the human living environment. Among those capabilities, mobility possesses the means of displacing itself in the most suitable manner. Portability is to be compact and lightweight, and preferably equipped with a self-contained energy source.

The use of robot technology is particularly desirable in applications in which would otherwise require to expose people to hazardous environments or which involve monotonous, exhausting, or dirty actions. Robots require high-power energy sources to supply consecutively for a long time because human can rarely deal with maintenance operations. Furthermore, to achieve a specific task, mobile robots are extensively installed with end-effecters that require high energy; for example, a vacuum cleaner, a manipulator or any other motor-driving tools.

Energy plays important roles. A service mobile robot requires a power system that can meet several goals simultaneously. A power source must store energy sufficient to allow the robot to perform a useful amount of work. Given the technology of today, however, a battery-driven robot cannot work for long on a single charge. Energy problems of mobile robots become a great challenge when mobility and portability have been taken into account while a mobile robot requires a high-efficiency power supply.

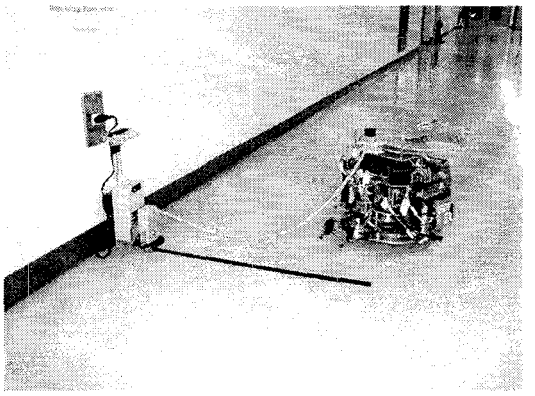


Figure 1. *HYPOS* Prototype System

In this paper, we propose *Hybrid Power Supply (HYPOS)* system that simultaneously uses the onboard battery and household electricity as power units for mobile robots. The purpose is to provide sufficient energy for a small, indoor service robot without losing its mobility.

We summarize the current technologies on the power system in Section 2 followed by the description of *HYPOS* technique. The development of a prototype system (Figure 1) and experiment on an useful task is presented in the following section. Finally we present some tentative conclusions and recommendations for further study.

2. Power Supply for Mobile Robots

Batteries are by far the most common solution employed by mobile robots for the problem of energy storage. The major advantage of batteries is that they are highly mobile. A conventional solution for the problems

0-7803-4300-x-5/98 $10.00 © 1998 IEEE

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on January 08,2026 at 12:47:27 UTC from IEEE Xplore. Restrictions apply.

of battery capacity is to let the robots return to power stations and plug-in for charging [2][3][4]. A significant disadvantage of charging is that it takes a very long time (several hours) for the battery to regain a full charge. New batteries are being launched into electric vehicles market but until now there is no commercially available, effective and economical electrochemical source for mobile robots [5].

The onboard power supply is not limited to the battery. Other energy storage techniques require mechanical means to produce electricity. For instance, AURORA [6] is powered by a 2600 [W] fueled onboard AC generator, which enables continued autonomous operation in agricultural areas. The robot platform, however, is rather large and its working area is limited to outdoor environment.

Mobile robots that work in a small room can be supplied by AC power through a power line from a fixed power station. This technique turns out to be useful for routine operations such as software development or program debugging. Yamaue et al. reported the development of a mobile cleaning robot with a 20 [m] cord fixed to an outlet [7]. K-CREITOR, a clean room investigation robot, was tethered to a supervising computer. A cable take-up machine was located off-board to assist cord-winding [8]. The disadvantages of using an electric cord are that the cord limits the robot's working area by its length and the cord tension may disturb the robot motion.

A new supply technology called "noncontact power supply system" has been introduced for specific applications. As for automated guided vehicle (AGV), an electric power transfer system through a fixed power transmission line by electromagnetic induction was proposed [9]. In the development of micro robotic system, the cordless power supply system was introduced to overcome the limitation of size and space on-board [10]. The limitation usage makes such technology not applicable to normal-size service robots.

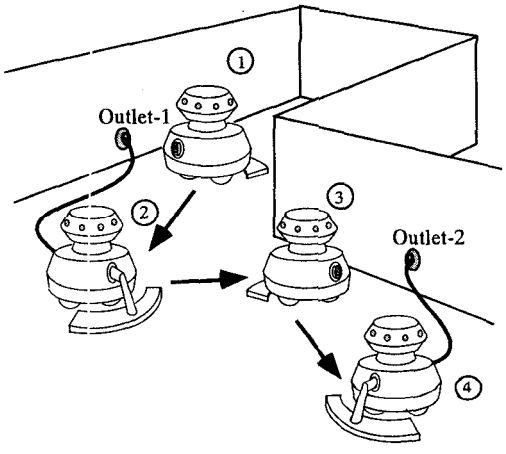


Figure 2. Image of *HYPOS* Concept in Cleaning Task

Solar cells can produce electricity from solar energy for free. It has been used as a power source of a solar car [11]. The famous Rocky 7 Mars Rover is also supplied by solar energy but its electronic device is well designed to consume low energy [12]. For common use, solar cells are too ineffective to use on anything but a huge, outdoor robot and as yet, solar cells have no power storage capacity.

It can be clearly seen that the current technologies on power supply can not meet the highly demand of service robots. In Section 3, we introduce a current technology-based methodology called *Hybrid Power Supply (HYPOS)* to break through this barrier.

3. Principle of *Hybrid Power Supply (HYPOS)*

We realize that to develop a new technology for energy storage may take numerous years. Our primary concept is to combine two or more numbers of existing methods to avoid their weak points and get a system, which performs better than those methods. In case of indoor tasks, the household electricity is considered as another useful source apart from a conventional battery.

Household electric utility provides powerful AC electricity. The channel for the electricity, an outlet, is extensively available in indoor environment. Power usually comes at steady voltage with low glitches and noises. Thus the operator is free from replacing the batteries and from worrying about operational time and recharging periods.

Batteries are required in a fully self-contained robot because there can be no power cord connecting the robot to the external supply. Batteries are highly mobile energy sources. The electricity is stored in chemical substance and can be produced without any special devices, a generator for instance. A robot with a battery is able to move around a large working space. It can go far away from the initial position as long as the battery can still supply energy to the system.

In this research we propose a hybrid technique called *HYbrid POwer Supply* (*HYPOS*) technique to integrate the function of the household electric utility and onboard batteries. This methodology takes both powerful energy and mobility into account to supply a mobile robot though, basically, both requirements can hardly meet. These two power sources are relatively adequate. Robot is highly mobile when powered from battery while it obtains high-power when consumes from the electric outlets.

As a result, a *HYPOS* robot is assigned to consume electric energy from the outlets while it needs powerful energy during the task operation. A battery powers the robot while it moves autonomously to search for and to connect to the outlets. To reduce the effect of the electric cord toward the movement, the robot should be equipped

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on January 08,2026 at 12:47:27 UTC from IEEE Xplore. Restrictions apply.

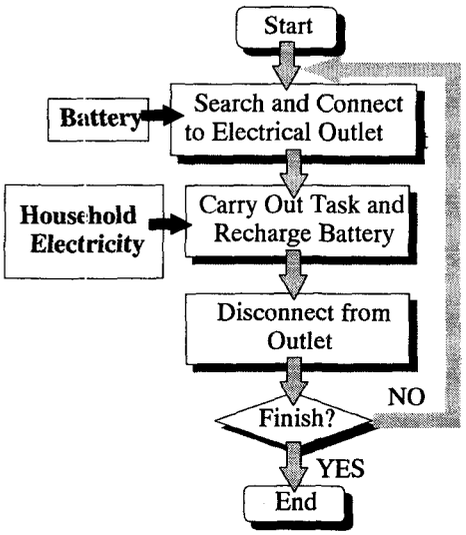


Figure 3. Flowchart of *HYPOS* Methodology

with a short cord and makes use of multiple electric outlets instead. More precisely, *HYPOS* technique supplies a mobile robot as follow (see Figure 2).

1. The robot starts looking for an electric outlet around the working area. Then the robot connects its cord to these outlets (Figure 2-1). These operations are completely autonomous and are powered by the batteries.
2. As the plug is already connected to the outlet, the robot switches to consume electricity from the outlet. The robot can work consecutively in a small area limited by the electric cord (Figure 2-2). In the same time, the onboard battery is being recharged.
3. When the task is complete or the robot cannot access the working area by the limitation of the cord, it disconnects the cord from the electric outlet and searches for other outlets for further operations (Figure 2-3 and Figure 2-4). The robot switches back to consume the energy from the onboard battery.

The *HYPOS* technique is summarized again in a flowchart in Figure 3. Each operation is executed repeatedly until the whole task is complete. A mobile robot with *HYPOS* system can work even when the battery is being recharged. *HYPOS* methodology also results in two attributes. First, the workspace is divided into several sub-areas to shorten the length of electric cord that may disturb the robot movement. Electricity from outlets is used in each sub-areas and those are connected by using battery. Second, because battery is functioned just for the movement around the outlet locations, the battery has been downsized and it does not take long hours to recharge.

HYPOS technique is applicable to mobile robots that work indoor by the fact that the electric outlets are extensively available. This technique is also effective for both robotic system that requires much energy for power-consuming appliances and the service robot that need to work consecutively for a long time.

We have realized that this methodology would be more effective if it requires little, if any, change in the infrastructure of the robot. The efficiency should be increased if the overhead time during searching and connecting to an outlet is decreased. In Section 4, the implementation issues based on the above objective are summarized.

4. Prototyping *HYPOS* Robot

Hybrid Power Supply (HYPOS) methodology has been realized on a real mobile robot. The primary function of a *HYPOS* robot is to find, to connect to, to disconnect from an electric outlet and to operate its electric cord autonomously. However, a number of restrictions on using the traditional outlets have been considered.

- A mobile robot, especially a small one, can not assert sufficient force to insert plug into the outlet. Even if it can, the wheel may slip and causes the positional error.
- The electrical outlet can be installed at any height above the floor because there is no standard for the installation. The exploration of electrical outlets is then the 3-dimension searching problem in space. That increases the complexity of plug-connecting task. Moreover, it is almost impossible to make use of the outlets installed higher above robot.

Instead of developing a robot with complete facilities, we let the environment provides some of them to simplify tasks. We developed a special device called *Power Supply Station (PSS)* attached to a conventional electric outlet to simplify the task of connecting to and disconnecting from the electric outlets. The custom-made plug on the robot can be easily connected to the *PSS* by electromagnetic force. And because the level of the connector above the floor is already defined, the task of finding *PSS* is reduced to 2-dimension searching problem.

4.1 Power Supply Station (*PSS*)

Power Supply Station (PSS) is an embedded device to a conventional electric outlet, as shown in Figure 4. The main components consist of an infrared beam emitter, an electric connector and a guideline on the floor. The infrared beam transmitter is the composition of 11 infrared LEDs arranging along the half-circular line and its driving circuit. They emit identical infrared beams into all directions in 180°, activating within 1 [m] radius, to indicate the location of *PSS*. The guideline is a 0.5 [m] line adhered on the floor. The guideline indicates the correct direction of a robot to connect to the *PSS*.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on January 08,2026 at 12:47:27 UTC from IEEE Xplore. Restrictions apply.

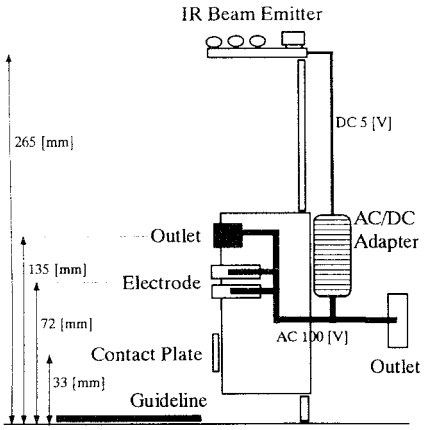


Figure 4. ***Power Supply Station (PSS)***

In summary, the infrared beams show the right "direction" to a *PSS* and the guideline indicates the correct "orientation" toward the *PSS* before the connection.

4.2 Prototype Robot

The basement robot is an omni mobile robot, 0.35 [m] in radius and 0.3 [m] high. The locomotion system is a 4-wheel synchro drive, which steers all wheels together at the same time. The whole unit is controlled by two V25 CPU boards. The onboard energy source is a 12 [V] 600 [mAh] lead-acid battery, weight about 700[g]. The battery powers the robot consecutively for 20 minutes on a single charge and it takes about 40 minutes to recharge. The schematic view of *HYPOS* robot is shown in Figure 5.

4.2.1 *PSS*-Search Devices

Conceptually, robot can reach a *PSS* by dead-reckoning system because the geometrical map of environment including the exact locations of *PSS*s is given. A set of special devices has been developed to establish the certain connection to *PSS*. *HYPOS* robot is equipped with an "infrared eye", an infrared detector, on the top-front panel. The robot can "see" and follow the beam of infrared from *PSS* within 1 [m] range. There are three line-tracing sensors installed under the robot platform. The line-tracing sensors are the infrared sensors, which are sensitive to the color of the detected object. In this case, the robot can distinguish between the black color of mark and rather light color of the floor. The orientation of robot toward *PSS* is examined through the execution of line-tracing procedure.

4.2.2 Cord-Winding Devices

An electric cord is essential for *HYPOS* robot to convey electricity from *PSS*. To perform autonomous operation, the cord must be located on the robot chassis itself and robot must wind the cord carefully to prevent over-tension or interference to the robot track. *HYPOS*

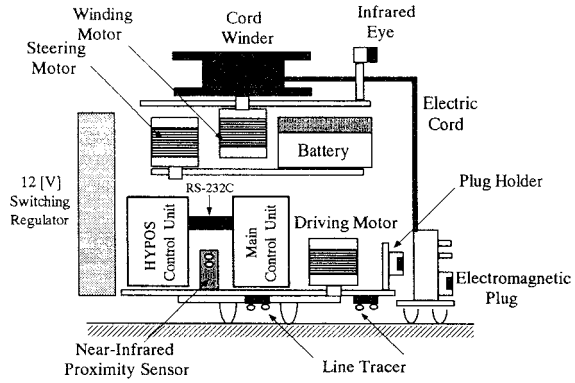


Figure 5. Schematic View of *HYPOS* Robot

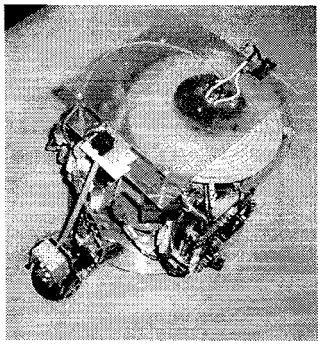


Figure 6. Cord-Winding System

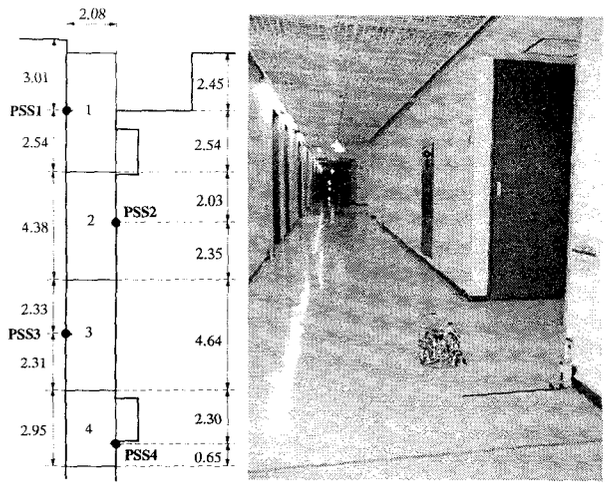
robot has a cord reel located on the top platform (Figure 6). There is an electromagnetic plug on the end of the cord. It consists of two electrodes, like the traditional one, with an electromagnetic holder. Instead of using the friction force between electrodes and outlet, the plug maintains connection to *PSS* by magnetic force.

The current control-scheme is that the winding operation is cooperated with the robot movement; paying out the cord as robot moving away from *PSS*, rewinding the cord as robot moving toward *PSS*. We do plan to add the potentiometer to measure the inclination angle of the cord that implies the cord tension to prevent over-tension and for smoother control.

5. An Experiment

This is a study of a specific task that has been chosen among variety of tasks done by indoor mobile robots. Sweeping is a popular task among the useful application of service robots. *HYPOS* methodology has been experimented in realization the sweeping task. The purpose of this experiment is to demonstrate how the *HYPOS* robot makes use of the *PSS* and to examine the efficiency of *HYPOS* technique in supplying an indoor robot. The experiment will be evaluated by the time that the robot can work consecutively. The evaluation is

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on January 08,2026 at 12:47:27 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Figure 7. Working Area (a) Map (b) Photo

compared to the traditional system that uses a lead-acid battery as energy storage.

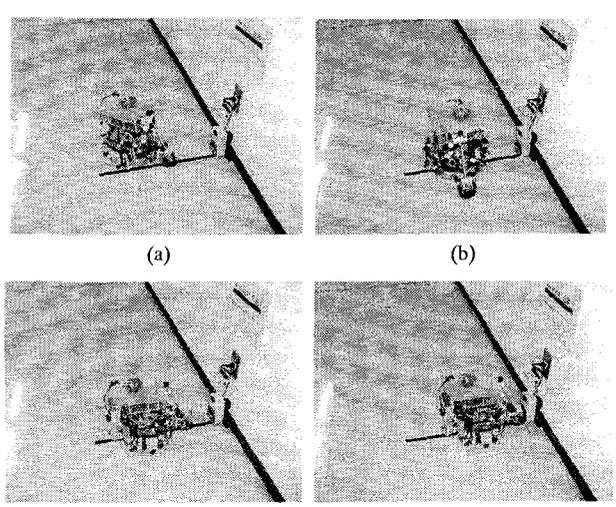
5.1 Condition of Experiment

The mission of our prototype *HYPOS* robot is to sweep the floor. Because there is no real sweeping tool installed on the robot, when the robot moves pass a 2-dimension area, that area is marked "cleaned". There is no obstacle in the working area. The parameter of the robot is summarized in Table 1.

The experiment is set up in the corridor of our laboratory with 4 *PSS*s as shown in Figure 7. *HYPOS* robot progresses the task by visiting *PSS*1-*PSS*4, repeating *PSS*3-*PSS*2 and finishing at *PSS*1.

5.2 Experimental Result

HYPOS robot can perform the task consecutively for 1 [hr] 45 [min] 56 [sec] and did finish the task. Figure 8 is the photos of *HYPOS* robot during the connection



(a) (b)

(c) (d)

Figure 8. *HYPOS* Robot connects to *PSS*-2

Velocity	0.03 [m/s]
Angular Velocity	π/6 [rad/s]
Sweeping Radius	0.20 [m]
Battery Life	25 [min]
Battery Recharge Time	40 [min]

Table 1. Parameters of *HYPOS* Robot

procedure to *PSS*-2. In Figure 8-(a), the robot approached *PSS*-2 and found the guideline on the floor. Figure 8-(b) and 8-(c), the robot corrected its orientation toward the *PSS*. The robot then moved forward along the guideline until it could connect to the *PSS* in Figure 8-(c). The raw data from the experiment is reported in Table 4 while the result graph classified by each operation is shown in Figure 9. The running time is divided into 4 phases; *Connect PSS, Sweep*, *Disconnect PSS* and *Move to next PSS*. The gray area in the graph is the productive time or the time that the robot uses for sweeping the floor. The overhead time is the time that loses while the robot moves between each *PSS* and connects to the *PSS*s. The efficiency of *HYPOS* technique in performing this sweeping task can be evaluated by the ratio of total productive time and overall operational time. Table 3 reports the efficiency of a lead-acid battery, which has been extensively used as an energy storage on mobile robots.

Total Operational Time	6,356 [sec]
Total Productive Time	4,681 [sec]
Efficiency	**73.6%**

Table 2. Efficiency of *HYPOS* system

Battery Life	25 [min]
Battery Recharge Time	40 [min]
Efficiency	**38.5%**

Table 3. Efficiency of Lead-Acid Battery

From Table 2, the efficiency of *HYPOS* system is 73.6%. Comparing to the data in Table 3, the efficiency of lead-acid battery is only 38.5 %. It is obviously that *HYPOS* system is more effective than the battery in term of uninterrupted operational time.

However, the electric cord effected the system a lot during the experiment. We found that the cord could not be paid smoothly because it often gets stuck in the winding spool. This sometimes caused the cord become over-tensioned that resulted in some trajectory error. The structure of cord winding system and the control scheme is subject to improve in future research.

6. Conclusions

This paper is the description of *Hybrid Power Supply (HYPOS)* for indoor service robots. *HYPOS* makes use of existing technology, including the conventional battery and the household electric utility, to power mobile robots.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on January 08,2026 at 12:47:27 UTC from IEEE Xplore. Restrictions apply.

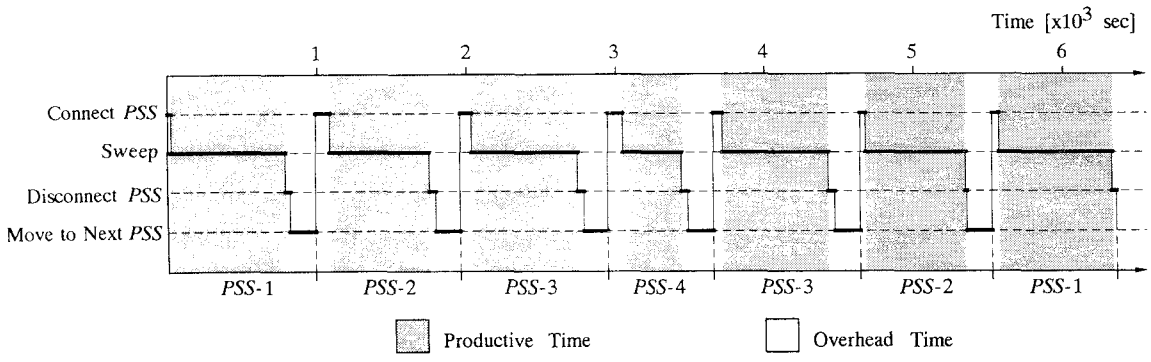


Figure 9. Experimental Result

Unit: [sec]	*PSS1*	*PSS2*	*PSS3*	*PSS4*	*PSS3*	*PSS2*	*PSS1*
Search & Connect to PSS	22	96	73	98	64	37	42
Sweep	763	665	712	399	710	672	760
Disconnect from PSS	31	44	46	42	45	11	30
Move to Next PSS	168	166	157	169	166	168	-

Total Time: 1 [hr] 45 [min] 56 [sec]

Table 4. Raw data from the Experiment

The methodology is applicable to a variety of indoor applications. The discussion of prototype design and the detail of developed robot are presented. The experimental result has shown that *HYPOS* system can supply a robot consecutively for almost 2 hours and its efficiency is 73.6% while the efficiency of a lead-acid battery is only 38.5%.

The work reported in this paper is work in progress. *HYPOS* robot's capacities are not complete. Especially, the cord winding module need to be more developed. The smooth winding of electric cable and the coordination between cord and robot motion will be carefully studied.

References

[1] S. Asami, "*Robots in Japan: Present and Future,*" IEEE Robotics & Automation Magazine, Vol.1, No.2, June 1994.

[2] M. Onishi and Y. Nishikawa, "*Autonomous Mobile Robot System for Transportation in the Clean Room,*" Robot, No.86, pp.33-37, May 1992.

[3] J. C. Wilson and P. A. Berardo, "*Automatic Inspection of Hazardous Materials by Mobile Robot,*" Proc. of 1995 IEEE Int. Conf. on Systems, Man and Cybernetics, pp.3280-3285, 1995.

[4] T. Hamada, N. Takehara, N. Oroku and K. Inoue, "*Application of a Mobile Clean Robot to Semiconductor Fabrication,*" Robot, No.109, pp.39-45, March 1996.

[5] M. J. Riezenman, "*The Great Battery Barrier,*" IEEE Spectrum, Vol.29, No.11, pp.97-101, November 1992.

[6] A. Mandow, J. M. Gomez-de-Gabriel, J. L. Martinez, V. F. Munoz, A. Ollero and A. Garcia-Cerezo, "*The Autonomous Mobile Robot AURORA for Greenhouse Operation,*" IEEE Robotics & Automation Magazine, Vol.3, No.4, pp.18-28, December 1996.

[7] Y. Yamaue, H. Arakawa, D. Takaoka and M. Yamada, "*Autonomous Mobile Robot for Cleaning,*" Prep. 7th Annual Conf. of Robotic Society of Japan, pp.445-446, 1989.

[8] J. Omomo and M. Maeda, "*Development of Clean Room Environment Investigating Robot,*" Robot, No.86, pp.27-32, May 1992.

[9] T. Tsumura, "*AGV in Japan,*" Proc. of 1994 IEEE/RSJ Int. Conf. on Intelligent Robots and Systems, pp.1477-1483, 1994.

[10] T. Fukuda and H. Ishihara, "*Micro Optical Robotic System with Cordless Optical Power Supply,*" Proc. of 1993 IEEE/RSJ Int. Conf. on Intelligent Robots and Systems, pp.1703-1708, 1993.

[11] Y. Iwatsubo and T. Takimoto, "*Next Stage for Solar Car,*" Journal of The Society of Instrument and Control Engineers, Vol.35, No.5, pp. 360-361, May 1996 (in Japanese).

[12] S. Hayati, R. Volpe, P. Backes, J. Balaram, R. Welch, R. Ivlev, G. Tharp, S. Peters, T. Ohm, R. Petras and S. Laubach, "*The Rocky 7 Rover: A Mars Sciencecraft Prototype,*" Proc. of 1997 IEEE Int. Conf. on Robotics and Automation, pp.2458-2464, 1997.

Authorized licensed use limited to: Guangdong Univ of Tech. Downloaded on January 08,2026 at 12:47:27 UTC from IEEE Xplore. Restrictions apply.